KSJ RESEARCH

Research Note Vol.4

Selling Power to a Data Centre

Equity in a hypothetical 100 MW solar asset — KEPCO contract (CfD), direct PPA with a data centre, or merchant: which pays for its risk?

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2026.10

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A direct PPA needs only KRW 5/kWh over the KEPCO contract — network charges, not price, decide the deal

Price evidence (official first) → cash flows and debt for three routes → break-evens (price, termination, network charges) → Monte Carlo → IC memo

Key metrics [*1,*2,*3,*4,*5]

Key metric	Value	p.
Hypothetical solar · CAPEX (KEEI LCOE basis)	100MW · KRW 175bn	1
Yr-1 output · share of data-centre use	140.2 GWh · 44%	1
Price corridor: KEPCO contract ~ industrial tariff	**147.7 ~ 181.9**	3
Equity IRR · CfD / PPA (mid) / merchant	**12.1% / 15.2% / 3.7%**	4
MOIC · CfD / PPA / merchant	3.93x / 4.24x / 1.66x	4
Gearing · CfD / PPA / merchant	80% / 75% / 38%	4
PPA break-even (no termination, IRR basis)	KRW 152.2/kWh	5
Required PPA price · premium (1% termination)	**152.7 · +5.0**	5
Ceiling on network charges	**KRW 29.2/kWh**	5
Monte Carlo · PPA equity IRR P10 / P50	11.6% / 14.9%	7
Monte Carlo · merchant loss probability (IRR < 0)	5%	7

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *3 KEEI Basic Research 25-04 (2025): solar installation costs — LCOE, capacity factor, WACC, unit cost *4 Bank of Korea ECOS KTB 3-yr, USD/KRW *5 KSJ Research
Note: KSJ Research. Asset and offtaker are hypothetical. Prices KRW/kWh, amounts KRW 100mn. Source grades and provisional inputs in A2–A4

15.2%

Equity IRR · direct PPA (corridor mid KRW 164.8/kWh)

Same plant, different buyer: equity IRR 3.7% → 12.1% → 15.2%

- KEPCO contract: auction cap KRW 147.7/kWh for 20 years → gearing 80%, equity IRR 12.1%
- Merchant: SMP KRW 113.2/kWh (12-month mainland average), no RECs for post-2027 entrants → 3.7%
- CAPEX matched to KEEI's 125.5 LCOE — at 2022 actual installed cost the KEPCO IRR would be 23.6%

+5

Required PPA premium (annual termination 1%, per kWh)

Price to accept a PPA instead of KEPCO: KRW 152.7/kWh (KRW 149.7/kWh with no termination)

- Rule: PPA price at which expected equity NPV (at the hurdle) equals the KEPCO contract; after termination the plant sells at SMP
- At 5% a year: +KRW 19 → KRW 166.4/kWh
- No public basis for the termination rate, so a 0–5% grid is shown; a real deal sets it from the offtaker's credit

KRW 29/kWh

Ceiling on network and ancillary charges the data centre can bear (per kWh)

KEPCO industrial KRW 181.9/kWh − required PPA KRW 152.7/kWh = room for network charges KRW 29.2/kWh

- A direct-PPA buyer pays the generator and, separately, KEPCO's network charges — above the tariff there is no reason to buy
- Network-charge methods are not public — the No.2 reason firms avoid PPAs (SFOC survey); KPX is studying a reform
- RE100 value (vs green premiums) raises the ceiling and tariff increases favour a fixed PPA — both left out to stay conservative

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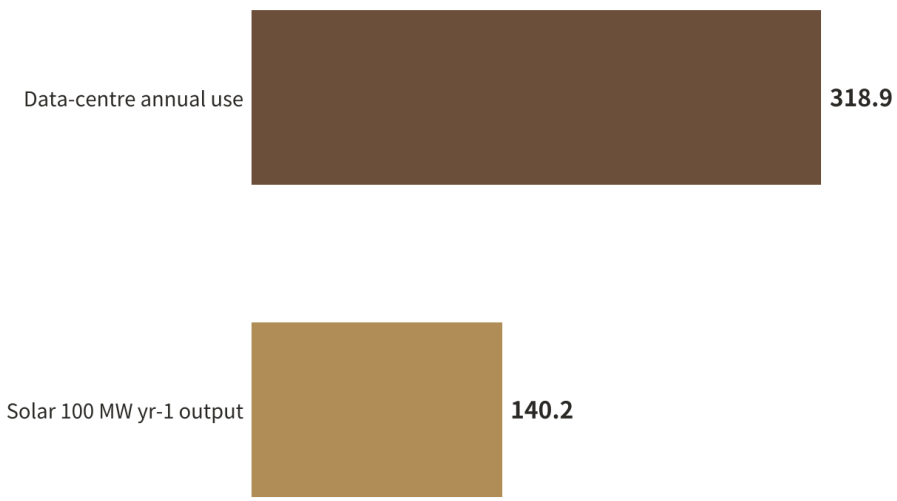
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Starting point: a 40 MW data centre's power bill is KRW 58.0bn a year — 100 MW of solar can cover 44%

Vol.2 data centre (lender's view) → Vol.4 the plant that sells it power (equity investor's view)

Data centre 318.9 GWh a year vs solar 140.2 GWh — 44% supplied directly [*1,*2]

Unit: GWh



Source: *1 KSJ Research Vol.2 (Oct 2026) *2 KSJ Research
Note: Vol.2 Korean case (IT 40 MW). Solar runs only by day, so the data centre still buys the rest from KEPCO

Power bill KRW 58.0bn a year — the cost Vol.2 passed to tenants [*1,*2,*3,*4]

Step	Value
IT capacity × load × PUE	40MW × 70% × 1.3
× 8,760 h = annual use	318.9 GWh
× KEPCO industrial tariff (2025)	KRW 181.9/kWh
= Annual power bill	**KRW 58.0bn**
Solar MW × capacity factor × 8,760 h	100MW × 16%
= Yr-1 output (0.5% annual degradation)	**140.2 GWh**

Source: *1 KSJ Research Vol.2 (Oct 2026) *2 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *3 NREL Jordan & Kurtz (2012), PV degradation review *4 KSJ Research
Note: Average industrial tariff (no tariff-class or time-of-use detail)

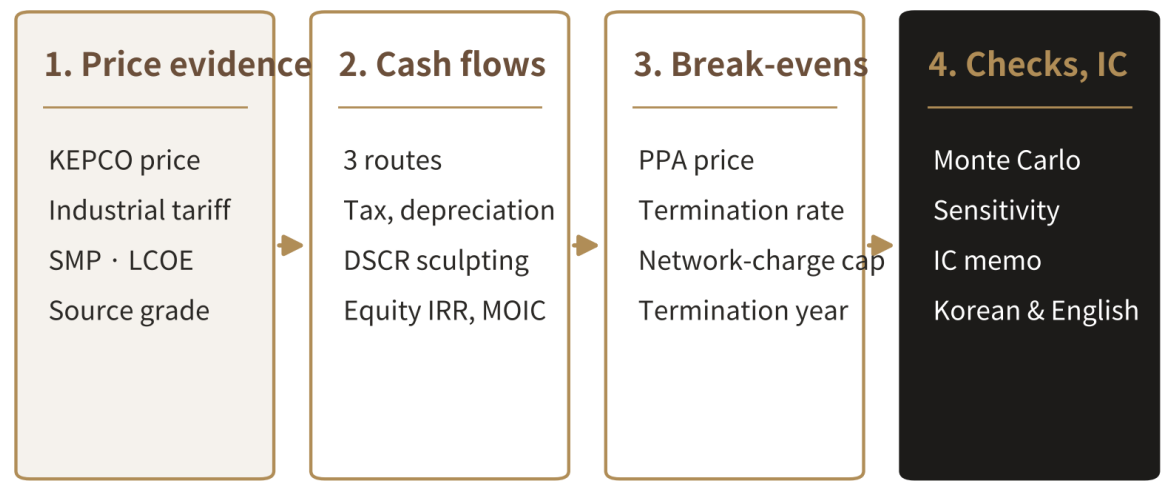
Question: if the plant sells to the data centre instead of KEPCO, does the extra price pay for the risk? [*1,*2]

An equity investor's three routes

- ① KEPCO contract (CfD): standard for new solar from 2027; 20-yr fixed price, KEPCO as counterparty
- ② Direct PPA: negotiated price; counterparty is Vol.2's data-centre SPV
- ③ Merchant (SMP): full price risk; no RECs for new entrants
- Vol.2 took the lender's view, Vol.3 the ECM view — this note takes the equity investor's

Source: *1 KSJ Research Vol.2 (Oct 2026) *2 KSJ Research

Structure: prices → cash flows → break-evens → checks and IC memo [*1]



Source: *1 KSJ Research

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Rules: from 2027 new solar defaults to a 20-yr KEPCO contract — a PPA is the alternative, merchant sales get no RECs

RPS repeal · contract market · distributed-energy cap · AI data-centre act → three routes

Timeline: from 2027 a KEPCO long-term contract is the default, a direct PPA the alternative [*1,*2,*3]

Date	Change	Grade
2026-02-03	Distributed-energy zones: firms on local grids to be allowed renewable PPAs; 35 MW cap under review	Press report of official release
2026-05-07	AI Data Centre Act passed — the requested LNG direct-PPA carve-out was dropped	Press report of official release
2026-05	Distributed resources capped at 40 MW in principle; a data-centre carve-out bill is pending (ministry opposed)	Press report of official release
2026-07-16	2026 round-1 solar auction: cap KRW 147.686/kWh (about 5% lower), ~1,000 MW	Official source
2026-09-15	Renewable Energy Act amended — RPS ends; a single long-term contract market with KEPCO	Press report of official release
2027-01-01	Effective — RECs restricted for new entrants; REC spot market until end-2029	Press report of official release

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Financial News (15 Sep 2026, citing the ministry): Renewable Energy Act reform *3 Herald K (9 May 2026) · Hankyung (3 Feb 2026): distributed energy, AI data-centre act
Note: Grades: official = government release checked · press report = media citing a government release (to be matched to the original, list in A4)

Three routes: price, counterparty and debt terms move together [*1,*2,*3]

Item	KEPCO contract (CfD)	Direct PPA with data centre	Merchant (SMP)
Price	Auction price fixed 20 yrs (cap 147.7)	Negotiated between KEPCO price and tariff	SMP (no RECs)
Counterparty	KEPCO	Data-centre SPV (Vol.2)	Market
Term	20 yrs	20 yrs (U.S. nuclear PPA precedents)	—
Target DSCR · max gearing	1.20 · 80%	1.30 · 75%	1.60 · 55%
Tenor	18 yrs	18 yrs	12 yrs
Equity risk	Capped upside · losing the auction	Offtaker credit · network charges · termination	All price risk

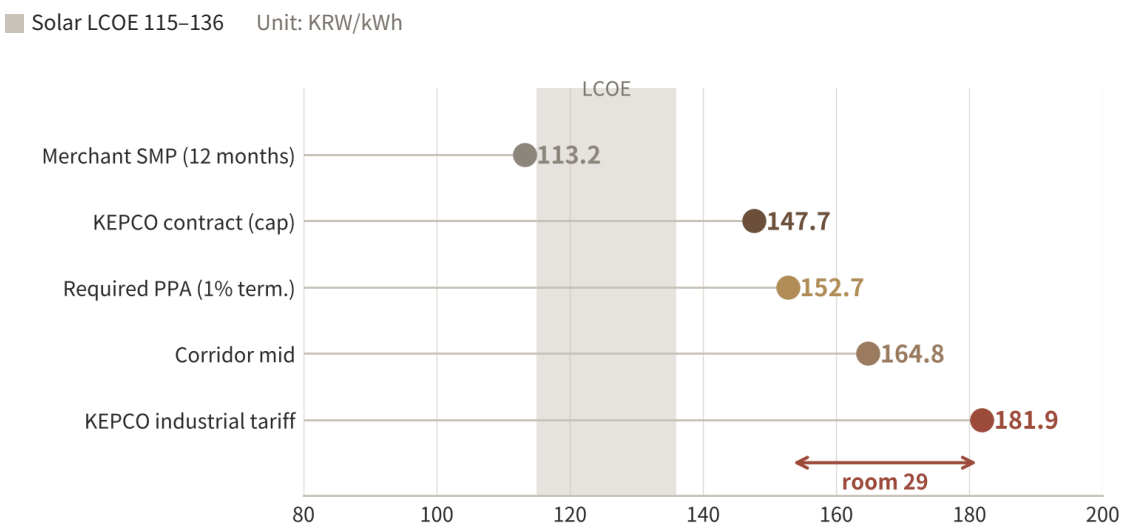
Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Financial News (15 Sep 2026, citing the ministry): Renewable Energy Act reform *3 KSJ Research
Note: Debt terms are author judgments (no public term sheets): weaker counterparty → higher DSCR, lower gearing. Rate = KTB 3-yr 3.84% (2026-10-20) + 150bp

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Price corridor: KEPCO contract 147.7 to industrial tariff 181.9 — 34.2 of room to negotiate a PPA

Floor = seller's alternative · ceiling = buyer's alternative · cost = LCOE · merchant = SMP · U.S. check

Corridor KRW 147.7–181.9: the 34.2 between is split by the PPA price and network charges *1,*2,*3,*4,*5



Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *3 KEEI Basic Research 25-04 (2025): solar installation costs — LCOE, capacity factor, WACC, unit cost *4 KPX EPSIS system marginal price (monthly weighted and hourly, mainland) *5 KSJ Research
Note: KEPCO price = 2026 round-1 cap (to be replaced by the KEA average award). LCOE = late-2024 study range

Price evidence: 4 of 4 from official originals — ministry, KEPCO, KPX, KEEI *1,*2,*3,*4

Item (KRW/kWh)	Value (range)	Grade	Published
KEPCO contract price (2026 round-1 cap)	147.7	Official source	2026-07-16
KEPCO industrial tariff (2025)	181.9	Official source	2026-02
Solar LCOE	125.5 (115~136)	Official source	2025
SMP base	113.2 (90~148)	Official source	2026-10

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *3 KEEI Basic Research 25-04 (2025): solar installation costs — LCOE, capacity factor, WACC, unit cost *4 KPX EPSIS system marginal price (monthly weighted and hourly, mainland)
Note: Tariff = 2025 industrial revenue ÷ volume · SMP = mean of monthly mainland SMP, Sep 2025–Aug 2026 (cross-checked with 10:00–16:00 hourly) · LCOE = 2024 Korean range

U.S. check: hyperscaler nuclear PPA at least $63/MWh — Korean solar contracts cost more *1,*2,*3

Item	USD/MWh	Grade
Talen–AWS nuclear PPA (17 yrs, 1,920 MW)	≥ 63	Official source
LevelTen N. America solar PPA avg (Q2 2026)	61.4	Press report of official release
Ref.: Korea KEPCO price (USD/KRW 1,332.1)	111	Official source
Ref.: Korea industrial tariff	137	Press report of official release

Source: *1 Talen Energy Form 8-K (11 Jun 2025, SEC EDGAR) *2 TaiyangNews (23 Jul 2026, citing LevelTen Q2 2026) *3 Bank of Korea ECOS KTB 3-yr, USD/KRW
Note: Talen: ~USD 18bn notional ÷ (1,920 MW × 8,760 h × 17 yrs) = average price at full, round-the-clock volume; the ramp to 2032 means the true price is higher (floor). Other nuclear PPAs (e.g. Meta–Constellation) are undisclosed

How to read it: the seller's floor is KEPCO's contract, the buyer's ceiling KEPCO's tariff *1

A PPA price lives between two outside options

- Seller: needs more than KEPCO (147.7) for extra risk → floor + risk premium
- Buyer: PPA + network charges must beat the tariff (181.9) → ceiling − network charges
- Merchant SMP (113.2) sits below the KEPCO price — no RECs for post-2027 entrants
- U.S. 24/7 nuclear at $63+/MWh — a Korean PPA is an RE100 premium product, not cheap power

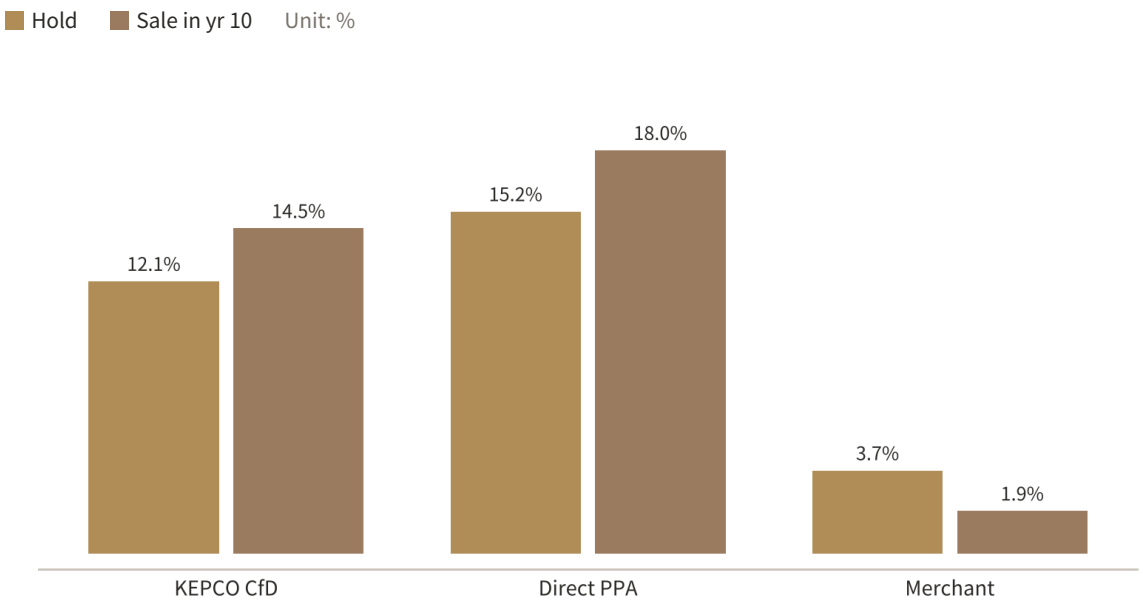
Source: *1 KSJ Research

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Equity IRR: PPA 15.2% · KEPCO 12.1% · merchant 3.7% — only merchant misses the hurdle

Debt sizing by route → equity IRR and MOIC → exit → sensitivity

Equity IRR: hold (25 yrs) vs sale in year 10 — the PPA leads either way [*1]



Source: *1 KSJ Research
Note: PPA at corridor mid KRW 164.8/kWh, no termination. Sale price = later equity flows at the buyer's 7%

Debt sizing: the stronger the counterparty, the more debt and the less equity [*1,*2,*3]

Route	Debt	Gearing	Min DSCR	Equity	Eq. IRR	MOIC
KEPCO CfD	KRW 140bn	80%	1.32x	KRW 34.9bn	12.1%	3.93x
Direct PPA	KRW 131bn	75%	1.56x	KRW 43.6bn	**15.2%**	4.24x
Merchant	KRW 66.6bn	38%	1.60x	KRW 108bn	3.7%	1.66x

Source: *1 Bank of Korea ECOS KTB 3-yr, USD/KRW *2 Corporate Tax Act art. 55 · Local Tax Act (law.go.kr) *3 KSJ Research
Note: CAPEX KRW 175bn (KRW 1,746k/kW, KEEI LCOE/WACC/CF basis) · progressive corporate tax + 10% local · straight-line 20 yrs · debt = min(DSCR sculpting, max gearing)

Sensitivity: CAPEX and capacity factor move IRR more than SMP — contracted routes carry cost risk [*1,*2,*3]

Input (low–high)	KEPCO CfD	Direct PPA	Merchant
CAPEX (KRW k/kW) 1,396–1,892	23.6% ~ 8.7%	25.8% ~ 12.0%	7.2% ~ 2.6%
Capacity factor 14%–18%	6.5% ~ 17.8%	10.0% ~ 20.5%	2.0% ~ 5.4%
SMP base 90–148	11.8% ~ 12.6%	15.1% ~ 15.5%	0.9% ~ 8.2%
OPEX (% of CAPEX) 1%–2%	14.4% ~ 9.8%	17.1% ~ 13.3%	4.7% ~ 2.7%

Source: *1 KEEI Basic Research 25-04 (2025): solar installation costs — LCOE, capacity factor, WACC, unit cost *2 KPX EPSIS system marginal price (monthly weighted and hourly, mainland) *3 KSJ Research
Note: One input at a time at its range ends (debt re-sized). CAPEX low = 2022 actual cost, high = matched to the 136 LCOE · CF range is an author sensitivity

How to read it: the KEPCO route earns through leverage, the PPA through price [*1]

Same plant, different capital structures

- KEPCO: price 147.7 vs LCOE 125.5, 80% debt → 12.1%
- PPA: 15.2% at the corridor mid — if nobody terminates (p5–p6)
- Merchant: lenders give only 38% and SMP is low → 3.7%
- Against the 8% hurdle: KEPCO CfD · Direct PPA clear, Merchant miss — between contracts it is price vs termination risk

Source: *1 KSJ Research

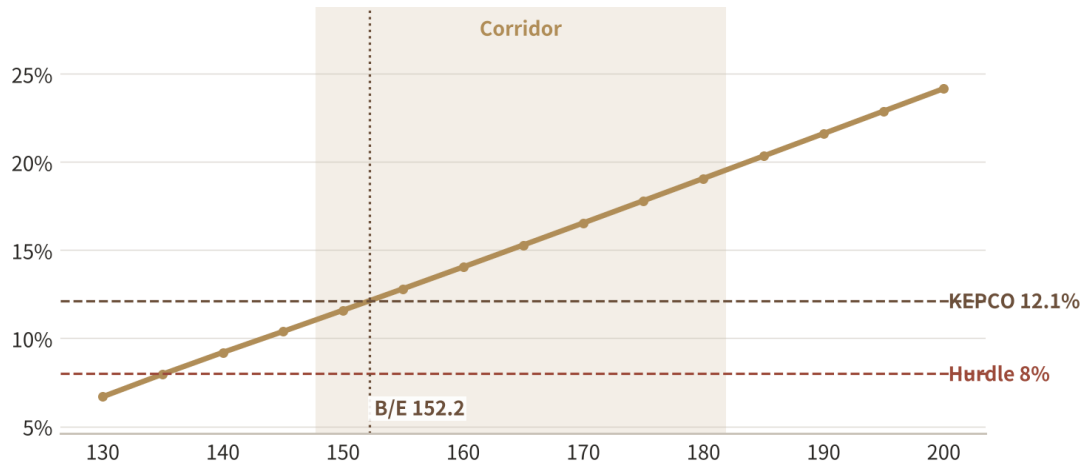
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PPA negotiation: at 1% termination KRW 152.7/kWh (+5) suffices — the data centre can bear 29 in network charges

Break-even PPA (IRR) → required price by termination rate (expected NPV) → network-charge ceiling

Equity IRR by PPA price: KRW 152.2 matches KEPCO — almost at the corridor floor *1,*2,*3

Unit: Equity IRR (no termination)



Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *3 KSJ Research
Note: Debt re-sized at each price (PPA terms). Break-even sits just above the KEPCO price because PPA debt terms are stricter

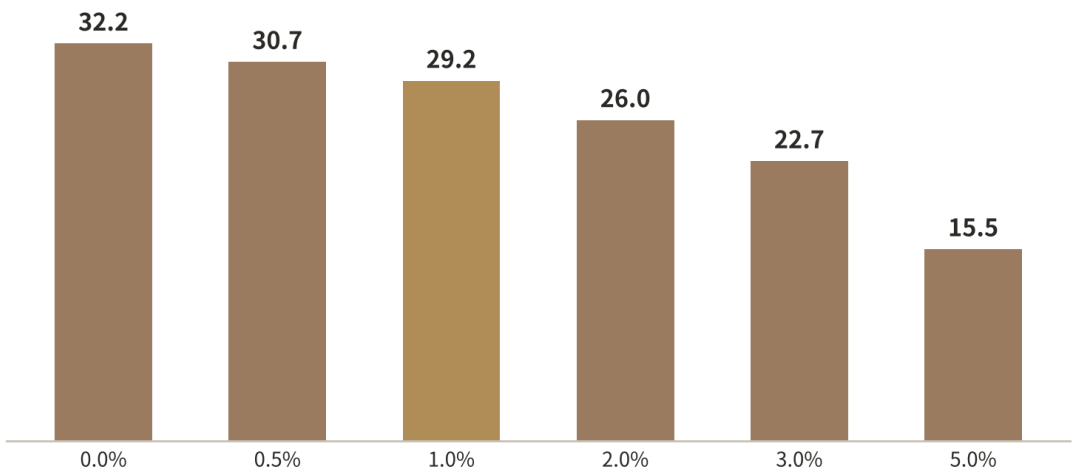
Required PPA price by termination rate: about KRW 5/kWh over KEPCO at 1% a year *1,*2,*3

Annual termination	Required PPA	Premium (KRW/kWh)	Network-charge cap (KRW/kWh)
0.0%	KRW 149.7/kWh	+2.0	32.2
0.5%	KRW 151.2/kWh	+3.5	30.7
1.0%	**KRW 152.7/kWh**	+5.0	**29.2**
2.0%	KRW 155.9/kWh	+8.2	26.0
3.0%	KRW 159.2/kWh	+11.6	22.7
5.0%	KRW 166.4/kWh	+18.7	15.5

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *3 KSJ Research
Note: Required price = PPA price at which expected equity NPV (at 8%) equals KEPCO's. Termination arrives at a constant annual rate during the PPA; afterwards the plant sells at SMP; debt keeps its no-termination sizing. Cap = industrial tariff − required price

Network-charge cap shrinks as termination risk rises *1,*2

Unit: KRW/kWh



Source: *1 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *2 KSJ Research
Note: x-axis = offtaker's annual termination rate; bar = the most the data centre can pay in network and ancillary charges on top of the PPA

The negotiation: price is easy to agree — the problem is that network charges are opaque *1

What the IC will ask

- Of the 34.2 corridor the seller needs +5 (1% termination); the other 29 covers network charges and buyer savings
- Opaque network-charge methods → if the buyer cannot see its ceiling, no deal
- Deal check: transmission/distribution tariffs, ancillary settlements, losses and night-time backup power in the buyer's model
- Seller's lever: termination payments and guarantees cut the effective termination rate and the premium (p6)

Source: *1 KSJ Research

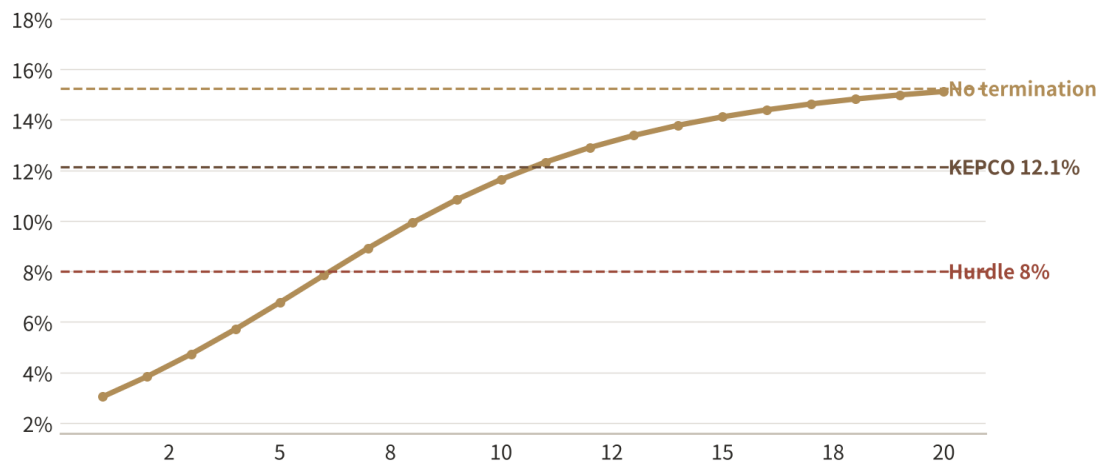
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Offtaker risk: a PPA that breaks within 11 years underperforms KEPCO — guarantees beat a premium

Equity IRR by termination year → hurdle-holding term → Vol.2 credit → contractual mitigants

Equity IRR by termination year: a break before year 11 does worse than KEPCO *1

Unit: Equity IRR



Source: *1 KSJ Research
Note: x-axis = termination year (SMP 113.2 from then). PPA at corridor mid KRW 164.8/kWh, debt fixed (no refinancing), no termination payment

To keep the 8% hurdle the PPA must last 7+ years *1

Termination	Equity IRR	vs KEPCO
Year 1	3.0%	below
Year 3	4.7%	below
Year 5	6.8%	below
Year 8	9.9%	below
Year 10	11.6%	below
Year 15	14.1%	above
Year 20	15.1%	above
No termination	15.2%	—

Source: *1 KSJ Research
Note: After termination the plant earns SMP, so early breaks hurt most — termination payments prop up the left side of this curve

Offtaker = Vol.2's data centre: lender DSCR in the 1.6x range, power passed to tenants *1

Item	Value
Data-centre SPV min DSCR (Vol.2 Korea)	1.63x
Data-centre equity IRR (Vol.2)	16.1%
Power cost treatment (Vol.2)	Billed to tenants (100%)
PPA volume / data-centre use	44%

Source: *1 KSJ Research Vol.2 (Oct 2026)
Note: The PPA is ultimately paid from tenants' power bills — if leases are shorter than the PPA, the gap becomes termination risk

Mitigants: contract terms that cut termination risk are cheaper than a premium *1

What to hold in the term sheet (A1)

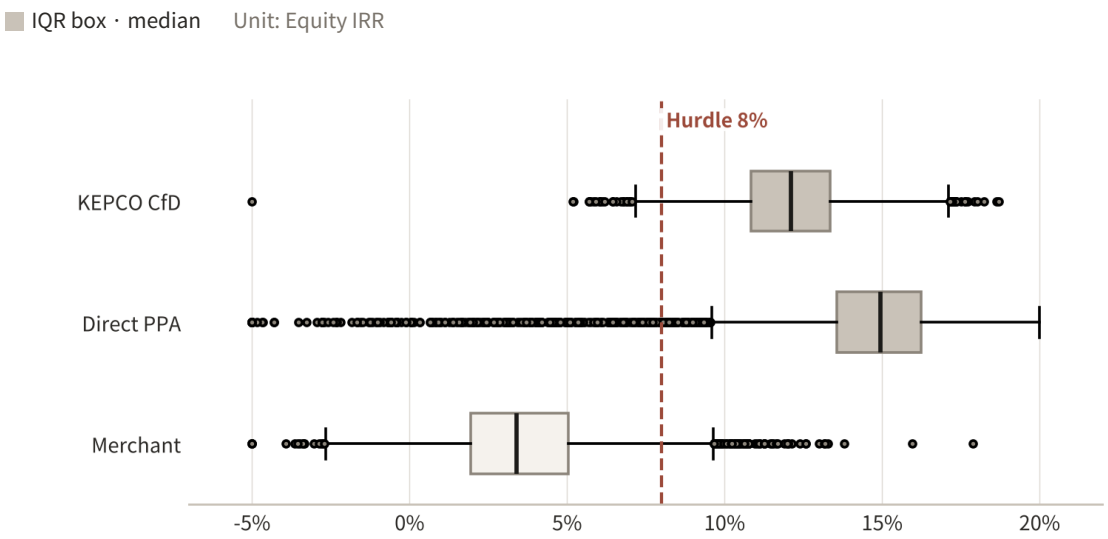
- Termination payment: covers part of the PPA-vs-SMP gap for the remaining term
- Credit support: parent guarantee or LC, or align lease power clauses with the PPA term
- Assignment: the PPA follows the data centre on sale or refinancing
- Change in law: a renegotiation rule if network-charge reform changes the buyer's cost

Source: *1 KSJ Research

Monte Carlo: PPA equity IRR P50 14.9% · P10 11.6% — merchant loses money 5% of the time

5,000 runs × 3 routes · seed 42 · SMP (log AR1) × output × PPA termination

Equity IRR distribution (5,000 runs): the PPA has a long left tail, merchant centres on 3.4% *1,*2



Source: *1 KPX EPSIS system marginal price (monthly weighted and hourly, mainland) *2 KSJ Research
Note: Common random numbers (SMP, output) across routes — differences come from the contracts only. PPA termination 1% a year. Shown −5% to 20%

Probability below the 8% hurdle: KEPCO 2% · PPA 5% *1

Route	P10	P50	P90	< hurdle	Loss (IRR<0)
KEPCO CfD	9.7%	12.1%	14.6%	2%	0%
Direct PPA	11.6%	**14.9%**	17.3%	5%	1%
Merchant	0.7%	3.4%	6.6%	96%	5%

Source: *1 KSJ Research
Note: Debt fixed at base sizing (equity funds shortfalls in bad years)

Inputs: SMP volatility 0.23 and mean reversion 0.72 estimated from 24 years of annual SMP (2002–2025) *1

Input	Value	Grade
SMP volatility	0.23	Official source
SMP mean reversion (AR1)	0.72	Official source
Annual output variation	5%	Author judgement
Resource estimate error	4%	Author judgement
PPA annual termination	1%	Author judgement
Runs · seed	5,000 · 42	—

Source: *1 KSJ Research
Note: SMP: log AR(1), mean-preserving · output: one resource error × annual noise · termination: geometric within the PPA term

How to read it: the PPA's risk is in the tail, not the average *1

What the distribution says

- KEPCO: narrow (P10 9.7%–P90 14.6%) — only output varies
- PPA: higher median but P10 11.6% — early terminations make the left tail
- Merchant: 5% loss probability — uncontracted new solar is not an equity case
- IC view: with a PPA, tail protection (termination payments, guarantees) comes before price

Source: *1 KSJ Research

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Implications: a direct PPA is about structure, not price — transparent network charges, termination protection, shape allocation first

Conclusion · IC checklist · limits · Vol.5 preview

Conclusion: a direct PPA is a structuring problem more than a pricing one *1

- Price: +5 over KEPCO (1% termination) is enough — well inside the 34 corridor
- Make-or-break: network charges must stay under 29/kWh — which needs a public method to judge
- Risk: early termination — termination payments, guarantees and assignment cut the tail and the premium
- Alternatives: KEPCO is safe at 12.1%, merchant earns 3.7%

Source: *1 KSJ Research

IC checklist: this note's assumption vs what a real deal must confirm *1

Item	This note	Confirm in a deal	p.
Price	Corridor 147.7–181.9	Average award (KEA) · tariff class	3
Network charges	Cap 29 (implied)	T&D tariffs · settlements · losses	5
Offtaker	Termination grid 0–5%	Rating · lease term · guarantees	6
Output	CF 16% (BNEF/KEEI)	P50/P90 yield report · curtailment	4, 7
CAPEX	KRW 175bn (LCOE basis)	EPC quote · land · grid connection	4
Shape	Pay-as-produced	24/7 firming cost (storage, KEPCO)	—

Source: *1 KSJ Research

Limits of this note *1

- CAPEX: 2022 actual installed cost (KRW 1,396k/kW) implies an LCOE of 100, below KEEI's 115–136 — matched to the LCOE to stay conservative
- PPA simplified to pay-as-produced — 24/7 shape risk and storage left out
- Tariff increases and RE100 value would raise the buyer's ceiling; excluded to stay conservative
- No public default data for termination — shown as a grid; a deal uses a credit rating

Source: *1 KSJ Research

Next, Vol.5 (finale): an AI energy asset investment map *1

Demand, equipment, PF, IPO and power on one page

- Place Vol.1–4 on one risk-return map — which asset fits which investor
- IC memo format: thesis, risks and mitigants, term sheet
- All assumptions and source grades in one table

Source: *1 KSJ Research

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IC memo: equity in a direct-PPA plant — price at the corridor mid, risks cut by contract terms

Summary · points · risks and mitigants · term sheet (illustrative)

Deal summary (illustrative — hypothetical asset) *1

Item	Detail
Asset	Hypothetical 100 MW solar, COD 2028
Revenue	20-yr direct PPA with a data centre, pay-as-produced
Uses · sources	KRW 175bn = debt KRW 131bn (75%) + equity KRW 43.6bn
Equity return	IRR 15.2% (corridor mid) · Monte Carlo P50 14.9%
Pricing rule	At least KRW 152.7/kWh (KEPCO +5, 1% termination)

Source: *1 KSJ Research

Three investment points *1

- Price edge: at the corridor mid, +3.1pp IRR over KEPCO
- Demand: AI data centres need RE100 supply — direct routes get scarcer after the 2027 RPS repeal
- Downside: on termination the plant still sells at SMP; debt service is checked with fixed debt

Source: *1 KSJ Research

Risks and mitigants *1

Risk	Mitigant
Offtaker termination	Termination payment · parent guarantee/LC · assignment
Network-charge opacity	Change-in-law reopener; shared buyer cost cap
Output shortfall	P90 debt sizing · curtailment compensation
SMP fall (after termination)	Re-contracting rights · re-bidding into the KEPCO market
Policy change	Check transition rules (pre-2027 commissioning)

Source: *1 KSJ Research

PPA term sheet (illustrative) *1

Clause	Proposal
Term	20 yrs (aligned with leases)
Price	KRW 164.8/kWh fixed (floor 152.7)
Volume	All output (pay-as-produced)
Termination	A share of the PV of (PPA − SMP) over the remaining term
Credit support	Parent guarantee or a one-year LC
Attributes	Renewable certificates to the buyer

Source: *1 KSJ Research
Note: Illustrative terms, not an actual contract

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Inputs and source grades (1/2): every number carries a grade

Value · range · unit · grade · source (links in the repository input file)

46 inputs: official 17 · press report 4 · secondary 0 · judgement 25 *1,*2,*3,*4,*5

Item	Value (range)	Unit	Grade	Source / basis
Capacity	100	MW	**Author judgement**	저자 설계값: 가상 태양광 100MW(AC). Vol.2 한국 데이터센터(IT 40MW) 연간 사용량의 약 4할을 공급하는 규모
COD year	2028	년	**Author judgement**	저자 설계값: 2026년 개발 착수 → 2028년 상업운전(신규 사업자는 2027-01-01부터 RPS 대신 계약시장 적용)
Installed cost per kW (2022 actual)	1,396 (1,396~1,635)	천원/kW	**Official source**	에너지경제연구원 기본연구 25-04 「재생에너지 보급비용 최소화를 위한 태양광 설치비용 영향요인 실증 연구」(공지영·장연재) PDF 93쪽: 국내 태양광 실제 설…
Capacity factor	0.16 (0.14~0.18)	비율	**Official source**	에너지경제연구원 기본연구 25-04 「재생에너지 보급비용 최소화를 위한 태양광 설치비용 영향요인 실증 연구」(공지영·장연재) PDF 10쪽 요약·31쪽 <표 2-…
Annual degradation	0.005 (0.003~0.008)	연 비율	**Official source**	NREL(미 국립재생에너지연구소) Jordan & Kurtz, Photovoltaic Degradation Rates — An Analytical Review(2…
Solar LCOE	125.5 (115~136)	원/kWh	**Official source**	에너지경제연구원 기본연구 25-04 「재생에너지 보급비용 최소화를 위한 태양광 설치비용 영향요인 실증 연구」(공지영·장연재) PDF 9·18쪽: 2024년 국내 …
WACC (LCOE cross-check)	0.06	연 비율	**Official source**	에너지경제연구원 기본연구 25-04 「재생에너지 보급비용 최소화를 위한 태양광 설치비용 영향요인 실증 연구」(공지영·장연재) PDF 10쪽·31쪽 <표 2-1>:…
OPEX (% of CAPEX)	0.015 (0.01~0.02)	CAPEX 대비 연 비율	**Author judgement**	저자 판단: 운영·유지비 + 보험(CAPEX 대비 연 1.5%). 에너지경제연구원 설치비 연구 원문에 운영비 수치 없음 — 공식 근거 미확보로 민감도(1~2%) 공개
OPEX escalation	0.02	연 비율	**Author judgement**	저자 판단: 물가 수준
Operating life	25	년	**Author judgement**	저자 판단: 태양광 모듈 보증 25년 수준
Depreciation life	20	년	**Author judgement**	저자 판단: 정액 20년 (법인세법 시행규칙 기준내용연수 확인 필요)
Corporate tax brackets	2:10,200:20,3000:22,inf:25	억원:%	**Official source**	법인세법 제55조(2026 사업연도부터) — Vol.2와 같음, 국가법령정보센터
Local income tax	0.1	법인세 대비	**Official source**	지방세법 — 법인지방소득세(법인세의 10%)
KEPCO contract price (2026 round-1 cap)	147.7	원/kWh	**Official source**	기후에너지환경부 보도자료 '2026년도 1차 태양광 경쟁입찰 공고'(2026-07-16): 입찰 상한가격 147.686원/kWh, 20년 고정가격. 낙찰 평균가는…
Contract term	20	년	Press report of official release	기후부 계약시장 개편(2026-09-15 공포, 2027-01-01 시행): 한전과 장기 고정가격 계약 — 파이낸셜뉴스 보도. 20년은 고정가격계약 관행
KEPCO industrial tariff (2025)	181.9	원/kWh	**Official source**	한전 전력통계월보 제566호(2025.12) 표 18·18-2: 2025.01~12 산업용 판매수입 50,971,210백만원 ÷ 판매전력량 280,221,380M…
SMP base	113.2 (90.4~148.4)	원/kWh	**Official source**	전력거래소 EPSIS 계통한계가격(가중평균 SMP, 월별 육지) 엑셀: 2025-09~2026-08 육지 SMP 12개월 평균 113.2원/kWh(범위 = 같은 …
SMP trend	0	연 비율	**Author judgement**	저자 판단: 명목 SMP 추세 0% (민감도로 공개)
SMP volatility	0.229	로그 표준편차(연)	**Official source**	전력거래소 EPSIS 계통한계가격(가중평균 SMP, 월별 육지) 엑셀: 2002~2025 연평균 SMP(24년) 로그 AR(1) 회귀 잔차 표준편차 0.229 (…
SMP mean reversion (AR1)	0.718	AR(1) 계수	**Official source**	전력거래소 EPSIS 계통한계가격(가중평균 SMP, 월별 육지) 엑셀: 같은 회귀의 AR(1) 계수 0.718 — 연료비 사이클로 평균 회귀
REC revenue	0	여부	Press report of official release	신규 사업자는 2027-01-01부터 REC 발급 제한, REC 현물시장은 2029년 말까지 — 파이낸셜뉴스(기후부 발표 인용). 2028년 상업운전 가정이라 R…
Annual output variation	0.05	비율	**Author judgement**	저자 판단: 연간 일사량 변동(해마다 독립)
Resource estimate error	0.04	비율	**Author judgement**	저자 판단: 장기 발전량 추정 오차(P50 오차)

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *3 KEEI Basic Research 25-04 (2025): solar installation costs — LCOE, capacity factor, WACC, unit cost *4 Talen Energy Form 8-K (11 Jun 2025, SEC EDGAR) *5 KSJ Research
Note: Grades: official (gold) = government, public body, statute or SEC original · press report = media citing an official release · secondary (red) = replace before final · judgement (brown) = stated basis, stress-tested

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Inputs and source grades (2/2): every number carries a grade

Value · range · unit · grade · source (links in the repository input file)

46 inputs: official 17 · press report 4 · secondary 0 · judgement 25 *1,*2,*3,*4,*5

Item	Value (range)	Unit	Grade	Source / basis
KTB 3-yr	3.84	%	Official source	한국은행 ECOS 국고채 3년 (기준일 값을 data/processed/ecos_ktb_daily.csv에서 읽음)
Credit margin	150	bp	Author judgement	저자 판단: Vol.2 한국 케이스와 같은 가산금리
Target DSCR · CfD	1.2	배	Author judgement	저자 판단: 정부 계약(한전 상대방) 매출 — 계약 매출 PF의 낮은 쪽
Target DSCR · PPA	1.3	배	Author judgement	저자 판단: 민간 오프테이커(데이터센터 SPV) 신용 위험만큼 높임
Target DSCR · merchant	1.6	배	Author judgement	저자 판단: 시장가 매출 — 가격 위험 반영
Max gearing · CfD	0.8	비율	Author judgement	저자 판단
Max gearing · PPA	0.75	비율	Author judgement	저자 판단
Max gearing · merchant	0.55	비율	Author judgement	저자 판단
Tenor · CfD	18	년	Author judgement	저자 판단: 계약 20년 안에서 2년 꼬리(tail)
Tenor · PPA	18	년	Author judgement	저자 판단: PPA 20년 안에서 2년 꼬리
Tenor · merchant	12	년	Author judgement	저자 판단
PPA term	20	년	Press report of official release	해외 하이퍼스케일러 원전 PPA 20년 사례(Meta–Constellation, Amazon–Constellation Calvert Cliffs) — 보도
PPA escalation	0	연 비율	Author judgement	저자 판단: 고정가(민감도로 공개)
Equity hurdle	0.08	연 비율	Author judgement	저자 판단: 지분 요구수익률(엑싯 가치 할인·손실 확률 기준)
Exit year	10	년	Author judgement	저자 판단: 인프라 펀드 보유기간
Buyer's required return	0.07	연 비율	Author judgement	저자 판단: 운영 단계 자산 매수자 요구 지분수익률(엑싯 가격 산정)
Data-centre IT capacity	40	MW	Official source	KSJ Research Vol.2 한국 케이스(1단계 20MW + 2단계 옵션 20MW)
PUE	1.3	배	Author judgement	KSJ Research Vol.2 가정(저자 판단)
IT load factor	0.7	비율	Author judgement	KSJ Research Vol.2 가정(저자 판단)
Talen–AWS PPA notional revenue	18	USD bn	Official source	Talen Energy 8-K 투자자 자료(2025-06-11) 3·7쪽: AWS PPA 17년 계약, 전량 기준 명목 매출 약 180억 달러(~$18bn not…
Talen–AWS term	17	년	Official source	Talen Energy 8-K 투자자 자료(2025-06-11) 3쪽: 17년 계약
Talen–AWS contract capacity	1,920	MW	Official source	Talen Energy 8-K 보도자료(2025-06-11): 전량 1,920MW, 2042년까지, 2032년까지 단계적 증가
LevelTen N. America solar PPA	61.4	USD/MWh	Press report of official release	LevelTen Energy 2026년 2분기 북미 태양광 PPA 평균 — TaiyangNews 보도(2026-07-23)

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *3 KEEI Basic Research 25-04 (2025): solar installation costs — LCOE, capacity factor, WACC, unit cost *4 Talen Energy Form 8-K (11 Jun 2025, SEC EDGAR) *5 KSJ Research
Note: Grades: official (gold) = government, public body, statute or SEC original · press report = media citing an official release · secondary (red) = replace before final · judgement (brown) = stated basis, stress-tested

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Data notes: official sources first; everything else flagged for replacement

As of 2026-10-20 · KTB 3-yr 3.84% (2026-10-20) · annual nominal model, KRW 100mn, KRW/kWh

3 inputs awaiting official sources — all replaced before final [*1]

Item	Current grade	Official source to use
Contract term	Press report of official release	기후에너지환경부 보도자료 원문(2026-09-15) 계약기간 확인
REC revenue	Press report of official release	기후에너지환경부 보도자료 원문
LevelTen N. America solar PPA	Press report of official release	LevelTen 원문 보도자료

> Built with an AI coding tool (Claude Code) for cash flows, debt sizing, break-evens, Monte Carlo and report generation; question design, source grading, assumption judgments and number checks by the author.

Source: *1 KSJ Research
Note: Rule: official originals first (government, public bodies, statutes, SEC) · media only when citing an official release · private data provisional only. Code and links: GitHub https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

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Glossary: the language of buying and selling power

Prices KRW/kWh · amounts KRW 100mn · U.S. USD/MWh

Glossary

Term	Definition
PPA	Power purchase agreement — seller and buyer fix price, term and volume directly
CfD	Contract for difference — settles contract vs market price; from 2027, long-term contracts with KEPCO
SMP	System marginal price — Korea's hourly wholesale power price
REC	Renewable energy certificate under RPS; restricted for new entrants from 2027
RPS	Renewable portfolio standard — replaced by a contract market in 2027
LCOE	Levelised cost of electricity — PV of lifetime costs ÷ PV of output
Network charges	Transmission and distribution charges a direct-PPA buyer pays KEPCO
Offtaker	The buyer under the contract (here the data-centre SPV)
DSCR	Debt service coverage = CFADS ÷ debt service
Sculpting	Shaping debt service so each year's DSCR equals the target
Equity IRR · MOIC	Return on equity · multiple of invested capital
Termination rate	Annual chance the PPA ends early (shown as a grid)
Price corridor	Range between the seller's and buyer's outside options
Pay-as-produced	Sell what is produced — no 24/7 obligation
Source grade	Official · press report of official release · secondary · judgement

Note: Shaded = core terms

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Disclosures

- This is a personal portfolio for education and research. It is not investment advice or a view on any asset or security.
- The 100 MW solar asset and the data-centre offtaker are hypothetical; no actual contract prices or terms are shown.
- Views are the author's own and do not represent any current or future employer.
- Every input carries a source grade (official · press report of official release · secondary · judgement); see Appendix A2–A4.
- As of 2026-10-20. Methods and code can be reproduced from the repository.
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https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

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